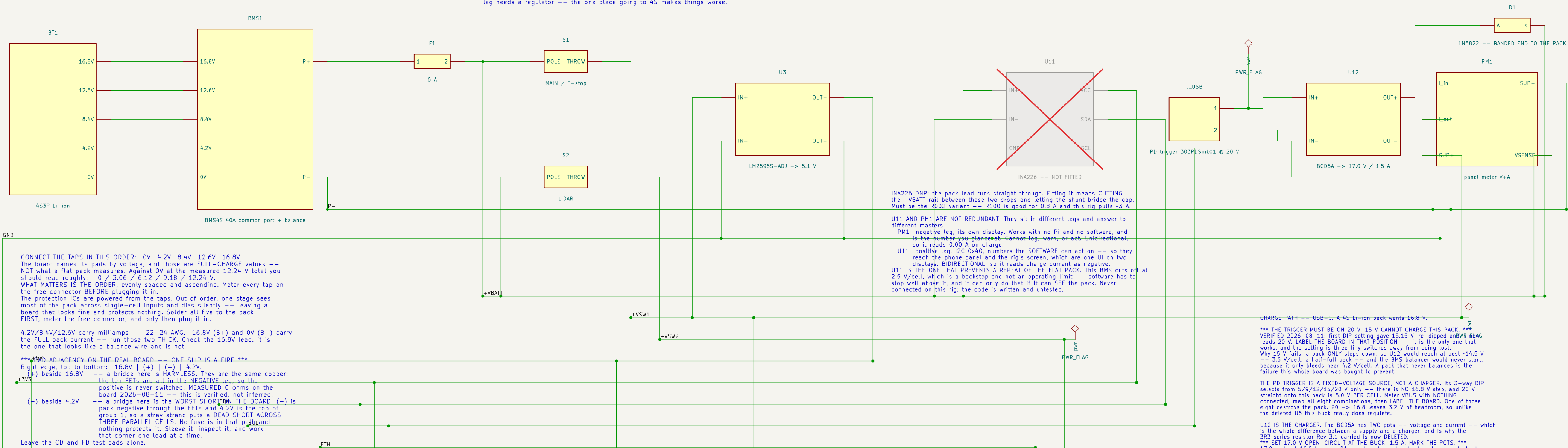


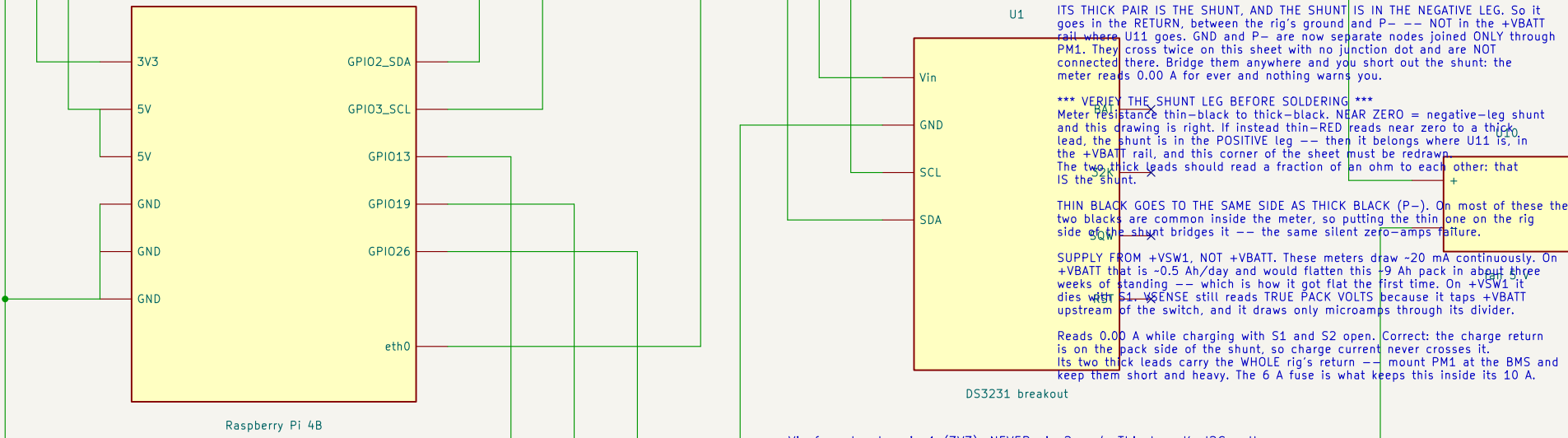
ZONE A POWER CHAIN 4S3P pack, common-port balancing BMS, USB-C charge

S1 and S2 both hang off the fused node, in PARALLEL. S1 does NOT switch the lidar --- opening S1 kills the Pi and the motor and leaves the VLP-16 spinning on S2.

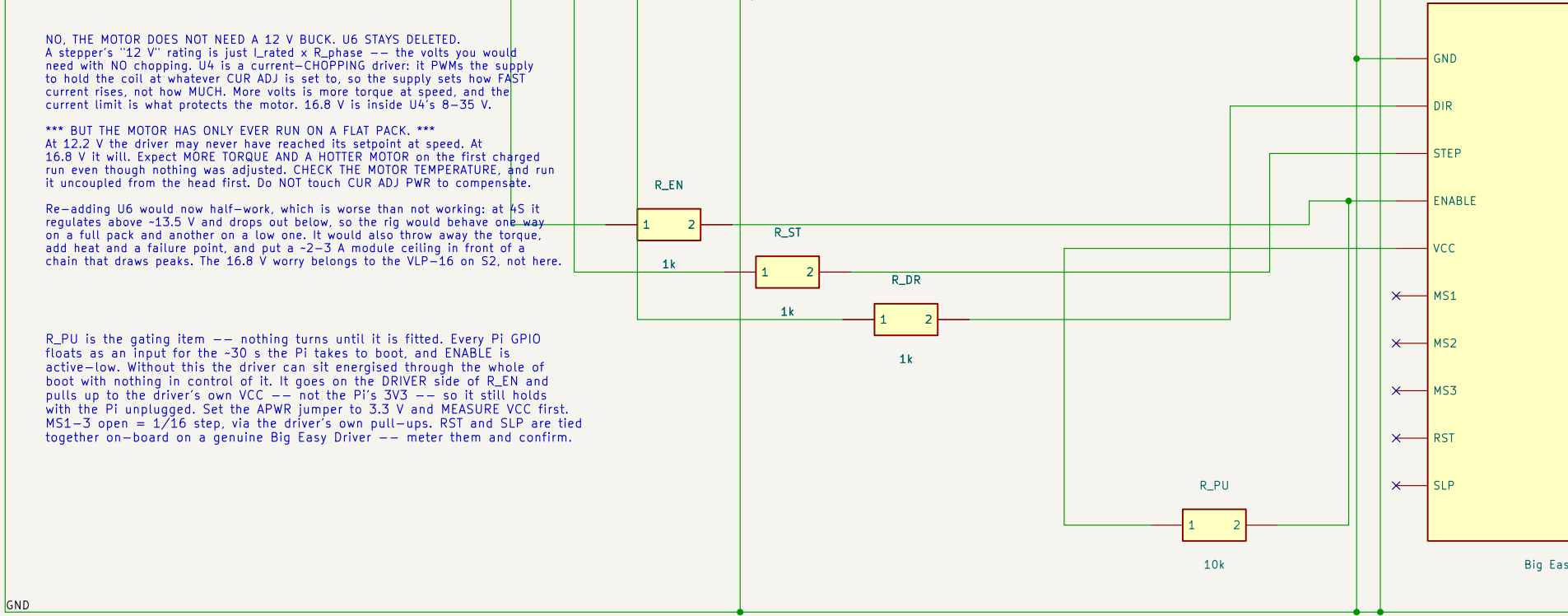
*** NEW ON 4S: S2 HANDS THE VLP-16 UP TO 16.8 V ***
On the old 3S assumption that leg never passed 12.6 V. CHECK THE SENSOR'S OWN INPUT RANGE before S2 is ever closed. If it will not take 16.8 V that leg needs a regulator --- the one place going to 4S makes things worse.



ZONE B THE PI AND ITS BUS



ZONE C MOTOR CHAIN



ACCEPTANCE TEST -- CHARGE FIRST, THEN MEASURE. The order is the whole point.

0. DONE 2026-08-11 --- U12 set on the bench to 16.8 V open-circuit and 1.5 A. NEITHER POT MAY BE TOUCHED AGAIN; mark them. With the trigger on 20 V there is a supply on this bench that would put 5.0 V/cell on the pack, and U12 is the only thing standing between them. EXPECT THE OUTPUT TO READ -12-13 V, NOT 16.8 V. THE MOMENT THE FLAT PACK IS CONNECTED. That is CC mode holding 1.5 A at whatever the pack sits at. It climbs to 16.8 V as the pack fills, then current tapers. Reading pack voltage instead of 16.8 V is the charger working, not a lost setting.
1. CHARGE AT 16.8 V. At 12.2 V this pack sits at 3.05 V/cell and the BMS is correctly latched on under-voltage: a common-port board releases when it sees charge voltage across P+/P-. Do this BEFORE judging the board.
2. Four groups --- 0V/4.2V, 4.2V/8.4V, 8.4V/12.6V, 12.6V/16.8V --- each -4.2 V when full and within 50 mV of the others. THE WEAK GROUP SHOWS UP HERE, and it is the one that tripped the cutoff. Leave it on charge an hour past 16.8 V: the balancer only works at the top, and that hour is the whole point of being a balancing power cell. B1/B2 the other. Cross them and the motor will see a balanced coil. If it is not, the motor will be unevenly loaded.
3. If the motor is not running, it is a sign of a bad setting. NO leg may have reversed direction and direction is a high-priority setting. NO leg may have P- with the motor and the motor will not run. The motor will be evenly loaded. If it is not, the motor will be unevenly loaded. The motor will be evenly loaded. If it is not, the motor will be unevenly loaded.
4. P+ to P- must equal 16.8V to 0V within a few mV.
5. Only then close S1, and only with the motor uncoupled from the head.

TLS Pie --- Rev 3.2 --- generated by kicad/make_kicad_schematic.py. Edit the script, not this file. Rev 3.0 assumed a 3S pack and was WRONG. Rev 3.2 fits the bought parts: BMS4S (COMMON PORT) --- no C- pad, so the old CHG- rail is deleted and the charge return is the star point) and the BCD5A CC/CV buck (two pots, so the 3R3 series resistor is deleted). Every conductor here is drawn: nothing is joined by name, and a crossing without a junction dot is not a connection. GPIO assignment is owned by tls.stepper.py